

HK IMO Training 2020-2021
Problem Set 20
For 19 June 2021



Problem 77. Any cell of a 5×5 table is coloured black or white. Find the greatest possible value of the ways to place a T -tetramino on the table so that it covers exactly two white and two black cells (for various colourings of the table).

Problem 78. Vulcan and Neptune play a turn-based game on an infinite grid of unit squares. Before the game starts, Neptune chooses a finite number of cells to be flooded. Vulcan is building a levee, which is a subset of unit edges of the grid (called walls) forming a connected, non-self-intersecting path or loop (*).

The game then begins with Vulcan moving first. On each of Vulcans turns, he may add up to three new walls to the levee (maintaining the conditions for the levee). On each of Neptunes turns, every cell which is adjacent to an already flooded cell and with no wall between them becomes flooded as well. Prove that Vulcan can always, in a finite number of turns, build the levee into a closed loop such that all flooded cells are contained in the interior of the loop, regardless of which cells Neptune initially floods.

* More formally, there must exist lattice points $A_0, A_1, \dots, A_k$, pairwise distinct except possibly $A_0 = A_k$, such that the set of walls is exactly $\{A_0A_1, A_1A_2, \dots, A_{k-1}A_k\}$. Once a wall is built it cannot be destroyed; in particular, if the levee is a closed loop (i.e. $A_0 = A_k$) then Vulcan cannot add more walls. Since each wall has length 1, the length of the levee is k .

Problem 79. Let N be a positive integer. Consider an $N \times N$ array of square unit cells. Two corner cells that lie on the same longest diagonal are coloured black, and the rest of the array is white. A move consists of choosing a row or a column and changing the colour of every cell in the chosen row or column.

What is the minimal number of additional cells that one has to colour black such that, after a finite number of moves, a completely black board can be reached?

Problem 80. A 2019×2019 square grid is made up of 2019^2 unit cells. Each cell is coloured either black or white. A colouring is called *balanced* if, within every square subgrid made up of k^2 cells for $1 \leq k \leq 2019$, the number of black cells differs from the number of white cells by at most one. How many different balanced colourings are there?

(Two colourings are different if there is at least one cell which is black in exactly one of them.)